

TiDAR: Short Summary + Throughput Snapshot

What TiDAR Does

TiDAR (Think in Diffusion, Talk in Autoregression) keeps a single decoder-only transformer but trains it to operate under two synchronized attention regimes:

- **Talk (AR)**: standard causal self-attention for verifying the current draft tokens.
- **Think (Diffusion)**: blockwise bidirectional attention for drafting the next candidate tokens.

Training doubles each sequence (clean + masked copy) so the diffusion branch predicts aligned targets with blockwise masks; inference uses structured masks to verify the current K tokens and pre-draft the next K candidates in a single forward pass.

One-Pass Inference Sketch

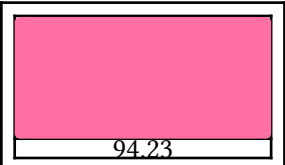
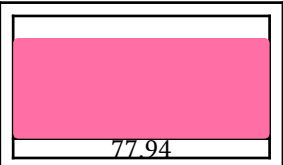
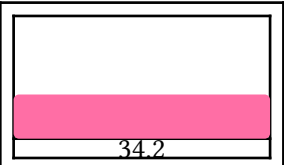
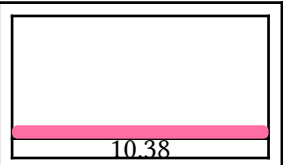
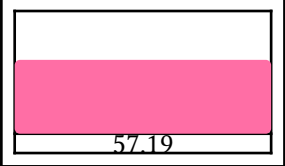
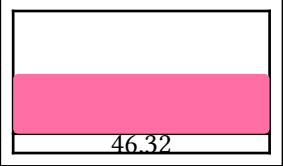
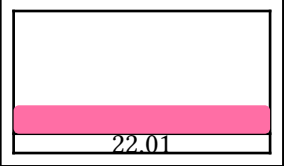
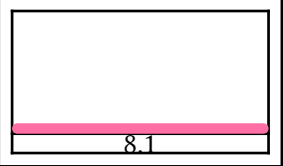
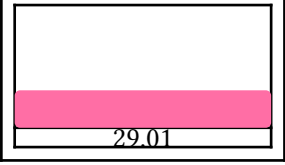
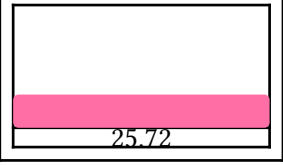
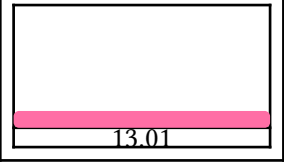
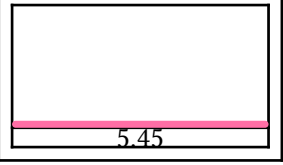
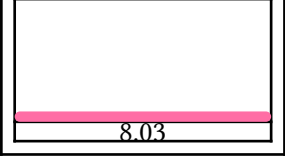
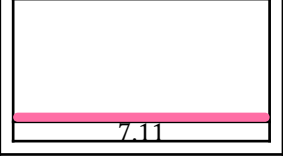
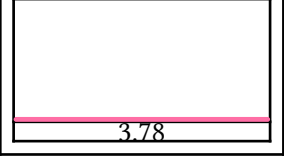
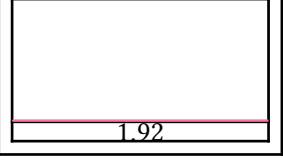
prefix tokens $\longrightarrow$ Talk (AR verify) $\longrightarrow$ commit r tokens

verify tokens (K) $\longrightarrow$

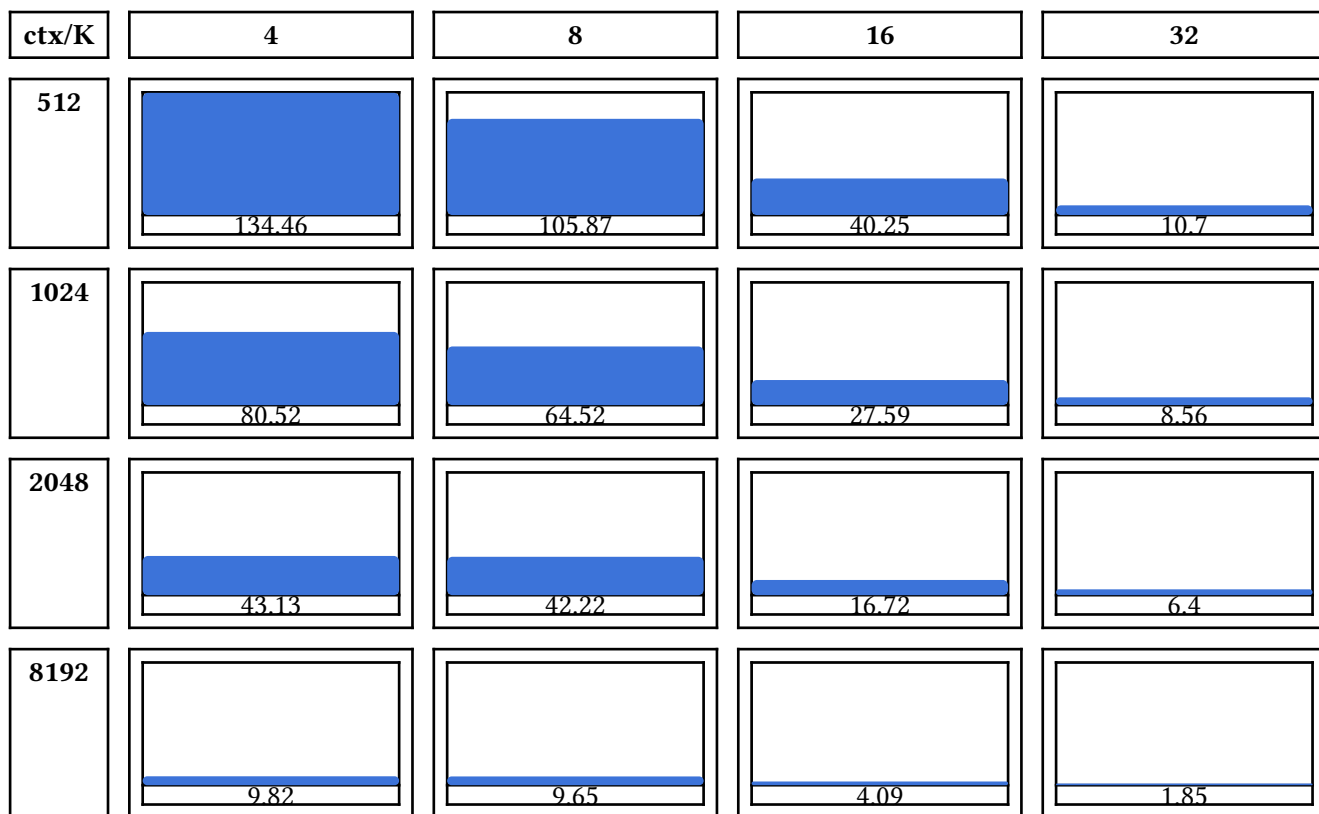
candidate blocks (K×K) $\rightarrow$ Think (Diffusion draft) $\rightarrow$ next K proposals

Throughput Sweep (RTX 3060)

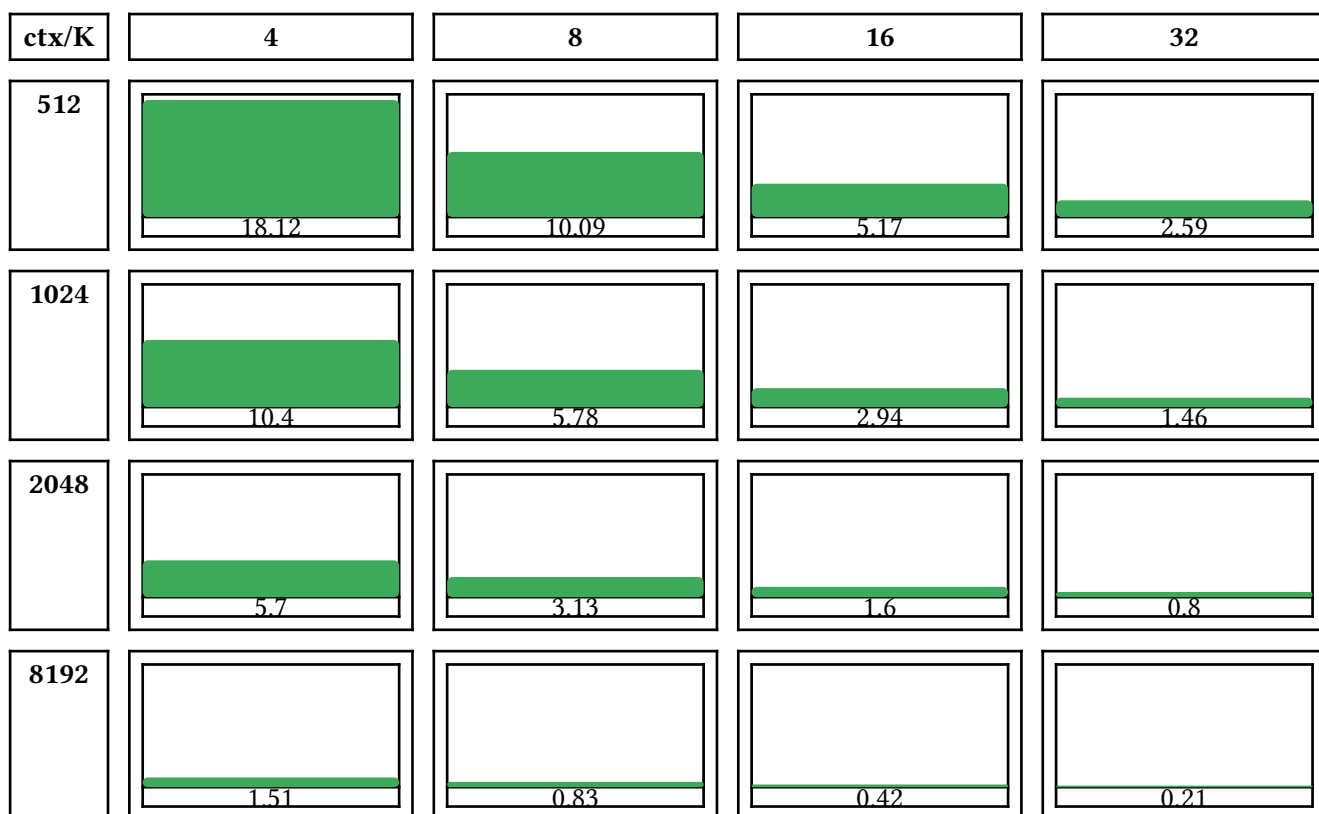
dense_bigmask_scan (iter/s)

ctx/K	4	8	16	32
512	 94.23	 77.94	 34.2	 10.38
1024	 57.19	 46.32	 22.01	 8.1
2048	 29.01	 25.72	 13.01	 5.45
8192	 8.03	 7.11	 3.78	 1.92

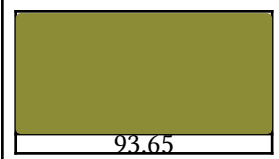
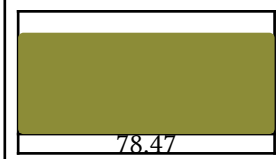
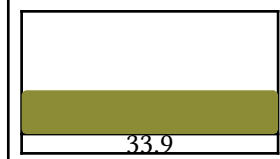
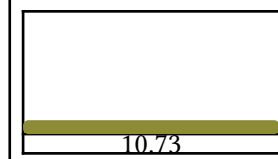
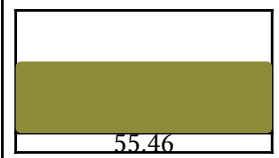
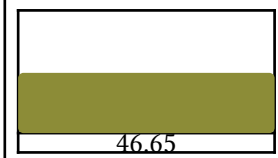
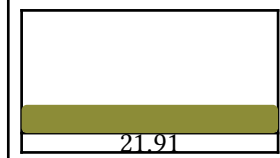
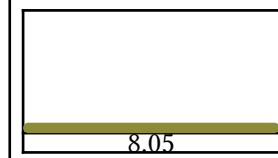
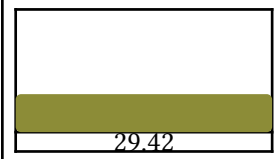
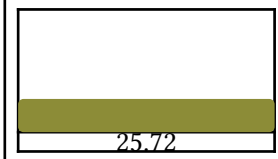
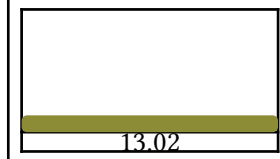
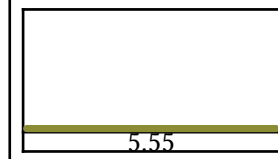
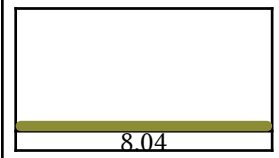
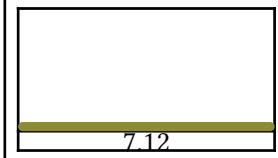
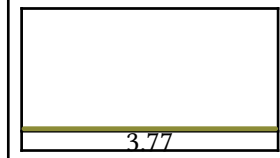
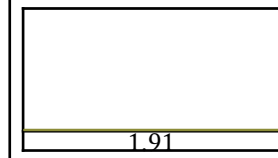
dense_tightkeys_scan (iter/s)



split_calls_scan (iter/s)



pallas_optional_scan (iter/s)

ctx/K	4	8	16	32
512	 93.65	 78.47	 33.9	 10.73
1024	 55.46	 46.65	 21.91	 8.05
2048	 29.42	 25.72	 13.02	 5.55
8192	 8.04	 7.12	 3.77	 1.91

Notes

- Metric: iter/s (1 / seconds_per_iter).
- Prefill length: round(0.66 * context_len).
- Config: heads=16, d=64, steps=128, iters=10, dtype=bf16, impl=cudnn.